

Is a passing bay worth a transporter?

Fleet-network co-design: the road-vehicle exchange rate (T6), a retrofit menu that includes control granularity (H4), and vehicle type under road occupancy (H7)

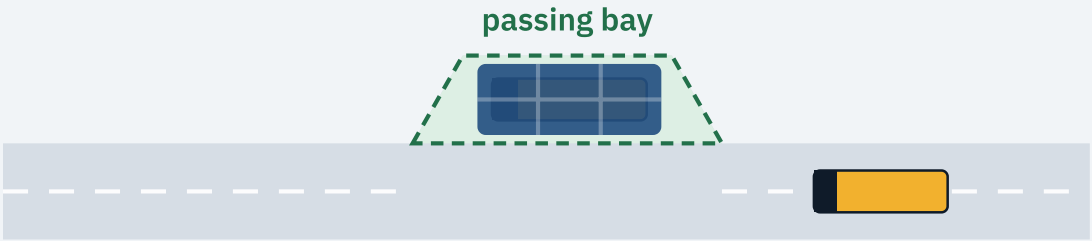
Is a passing bay worth a transporter? Fleet-network co-design for road-occupying transport (working title)

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Research manifesto v1: C4, H4, H7, T6 · T22–T24 established the capacity intervals, the system bound and two columns of dual prices; T27 is running, and T25 builds the first exchange-rate table after it

exchange rate = dual price ÷ daily capacity per vehicle



On a single lane, the loaded carrier pulls into the bay and an empty one passes

≈ ?



how many vehicles is it worth?



The sub-study in one minute

1

The question

To expand capacity where the network binds: buy vehicles or fix roads, large vehicles or small? We price each road change in vehicles.

2

The dividing rule

Plateau / T1' near 1: the bound is tight, its dual prices are the exchange rate, and fixing the road pays. Far below 1: the rule is the bottleneck, so change the orchestration first.

3

Results so far

Over 10 seeds: segmenting Okpo's dock road lifts the upper edge T1' by +8.0% but the plateau by only +0.3%. In the erection mix the dock binds (0.87): +76.5% and +8.3%, and the ratio falls to 0.53.

4

Next

T23 delivered two columns of dual prices and the route slack; T27, a less conservative deadlock-free rule, is running; then T25 builds the first exchange-rate table and T26 adds floating docks.

Where the dock mouth dominates, a road that can be used in sections may be worth more than a wider one.

Chapter 5's question: when we spend to expand capacity, vehicles or roads, large vehicles or small?

When loaded vehicles hold the road exclusively, what determines the capacity of a shipyard block-transport system? How should fleet, orchestration and network be designed together?



What the chapter takes in

- **Chapter 3:** the bound $T1'$ and its noise-free system version, the route-free $T1''$, and two columns of dual prices ($T23$)
- **Chapter 4:** the best deadlock-free rule, the interval's lower edge and the inner evaluator; now whole-route reservation ($T22$), re-checked in $T27$
- **MY-B:** vehicle types, the team-transport model and vehicle prices from 33 tender awards

What the chapter delivers

- The $C3/C4$ dividing rule and the first road–vehicle exchange-rate table
- A proof of $T6$ and critical-element identification; a retrofit menu with control granularity
- A bilevel co-design algorithm; the vehicle-type result ($H7$); upgrade values handed to chapter 6 for cross-yard prediction

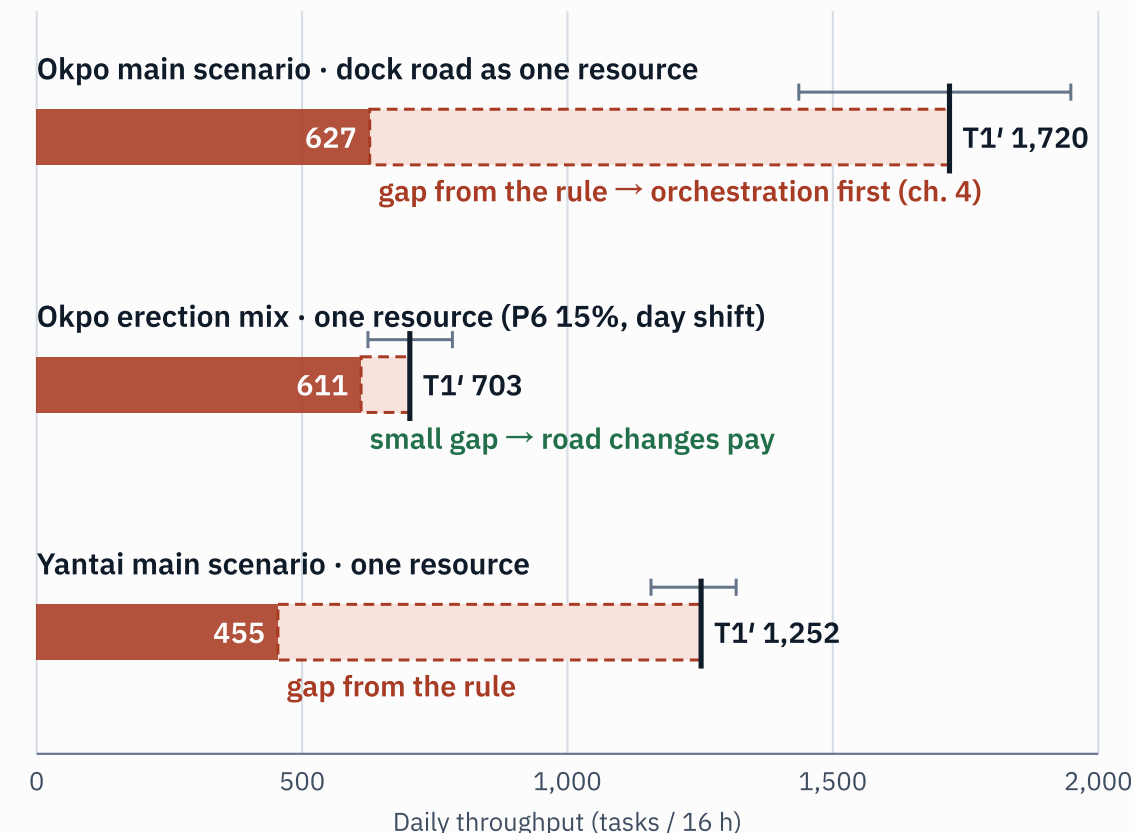
The manifesto items this chapter serves: C4, H4, H7 and theorem T6

#	MANIFESTO TEXT (V1, 9 OCT 2026)	HOW THE CHAPTER DELIVERS IT
C4	Road–vehicle exchange rate: where the bound is tight, upgrading critical elements can replace vehicles, at a rate equal to the dual price converted to vehicles. When the plateau is far below the bound, change the orchestration first. This is the boundary between C3 and C4	The dividing rule is the chapter's backbone; T22's 10 seeds give two examples (main scenario, erection mix)
H4	Hypothesis: where the network binds, one upgrade on a critical element is worth at least one vehicle; critical elements can be identified by dual prices. Upgrades to choose from: segmented blocking, dock-front holding bays, passing bays, one-way loops, widening, a second entrance. Criterion: the difference in vehicles needed before and after the upgrade in simulation, compared with the dual-price prediction. The two are expected to agree only when plateau / bound is close to 1. Holds: the exchange rate holds. Fails: the capacity bottleneck is orchestration, and chapter 4 is strengthened.	T25 (after T27): 10 seeds, full K grid; difference in vehicles needed at whole-yard volume; dual predictions from T23's two columns
H7	Hypothesis: large vehicles occupy the road both loaded and empty; small ones can yield when empty, but teaming up for a lift costs extra empty runs. Case: multi-vehicle transport of giant blocks for floating docks. Holds: the optimal fleet mix changes with how strongly the network binds. Fails: vehicle type and network can be decided separately.	T26 floating-dock scenario (Okpo rev 1833, in use from T27 step 0); MY-B vehicle types and prices
T6	Road–vehicle exchange rate: given by the dual prices; only upgrades on binding constraints have value. For new links, use T1" Status: author	Proved by the author; T23 has supplied T1" (draft proof, rewritten by the author) and two columns of dual prices

C3 (the other side of the boundary): where the network binds, how much of the bound is reachable is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks.

Source: research manifesto v1 (thesis-core, tag manifesto-v1, docs/研究宣言.md), §§2–4, quoted; only the author changes the wording. The right column describes how we carry it out (task board T22–T27; T22 and T23 reports at thesis-core a882742).

The C3/C4 boundary: how far the plateau sits below the bound decides between orchestration and road changes



■ Whole-route reservation plateau (mean of K = 100–150) ■ Gap to T1'

■ T1' (whisker: range over 10 seeds)

Sources: thesis-core a882742, docs/T22_report.md §4; plateaus and T1' read by make_data.py from the curves.csv of the batches listed there (saturated demand, 4 days, 10 seeds; plateau = mean of whole-route reservation over K = 100–150; whisker = the 10-seed range of T1', from T23's system.csv); completed-mix caliber from docs/T23_report.md §4.4. Okpo derived map rev 1825, Yantai rev 396 (Okpo moves to rev 1833 from 20_数据/船厂地图 at T27 step 0). Erection mix = P6 at 15%, day shift. Rule: review §2.3; manifesto C1, C4.

36%

36% Far below 1: the rule is the bottleneck

Raising the bound does not move throughput; change orchestration first, as chapter 4 pushes the lower edge up. Example: segmenting road 161 in Okpo's main scenario lifts T1' by +8.0% but the reservation plateau by only +0.3%; Yantai's main scenario also sits at 36%.

87%

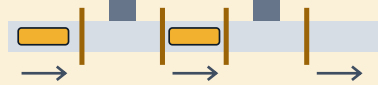
87% Near 1: the bound is tight

Dual prices are the exchange rate; upgrading the binding element pays, so fix the road first. Example: in the erection mix the dock road 161 binds (86% on the completed-mix caliber), and segmenting it raises the reservation plateau by +8.3%.

36%

Capacity is an interval [best deadlock-free rule, T1']: the lower edge is whole-route reservation (the best of the rules T22 tested; T27 re-checks). Chapter 4 pushes it up, chapter 3's tighter bounds press the upper edge down, and their ratio decides where chapter 5's money goes.

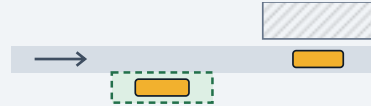
Retrofit menu: six changes, one purely in control granularity



CONTROL GRANULARITY

Segmented blocking (with direction locks)

A long road is cut at its stops into several resources; vehicles may follow in the same direction. Only the resource division changes, not the routes, so T1' still applies.



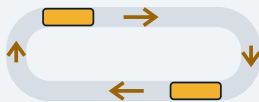
Dock-front holding bay

Vehicles waiting for the crane stand off the road and stop blocking the way to neighbouring stops. Where vehicles wait is not in the LP, so the value is read from simulation.



Passing bay

A wider stretch on a single lane where two vehicles pass. It changes routes, so the route-free T1'' prices it.



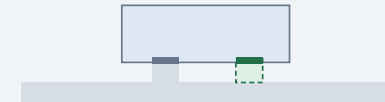
One-way loop

Turns head-on conflicts into same-direction following. Directions and shortest paths change, so T1'' is needed.



Widening

Raises width capacity, in steps: 1 m more on the 9 m road 161 leaves the bound unchanged; at 12 m two of the narrowest vehicles pass (T23).



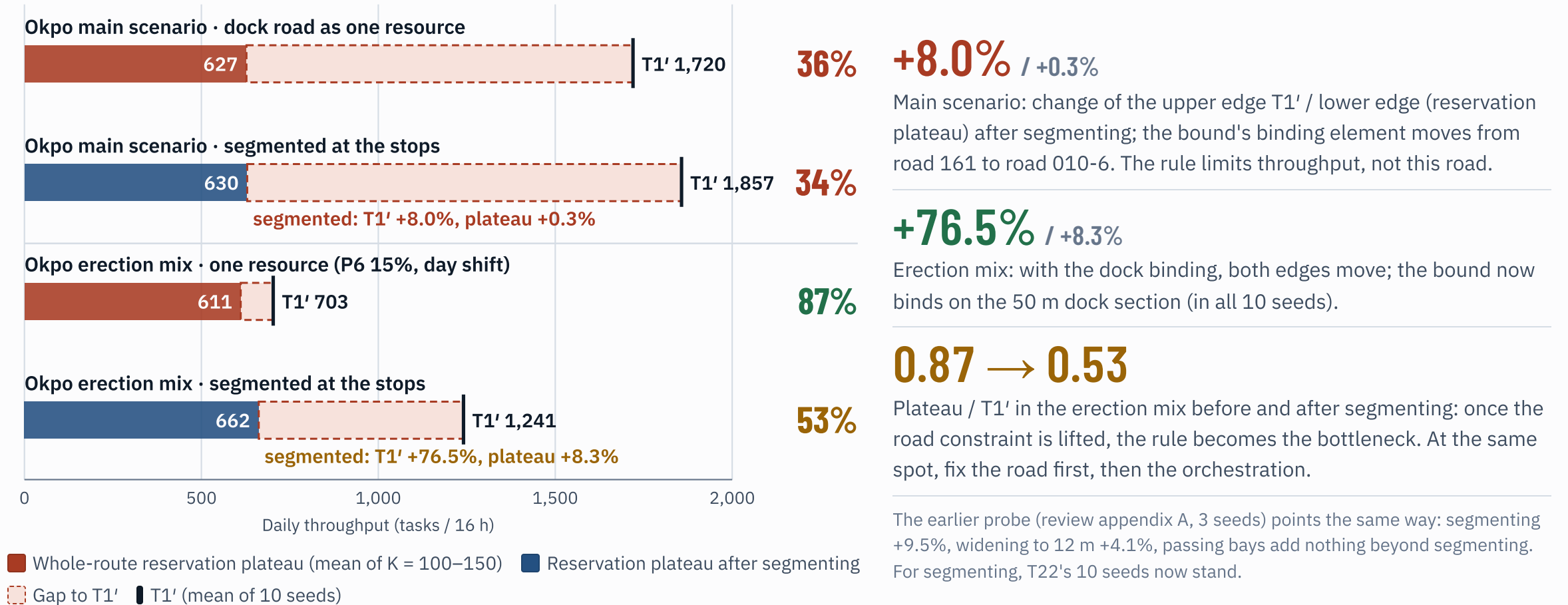
Second entrance

Adds a door where a door binds, sharing out the loading time held at the node. It adds a link, so T1'' is needed.

Control granularity is hidden capacity: segmented blocking costs no construction; it changes only how resources are divided and how vehicles are released. The manifesto lists block granularity under C3's orchestration and segmented blocking in H4's menu, so it sits right on the boundary.

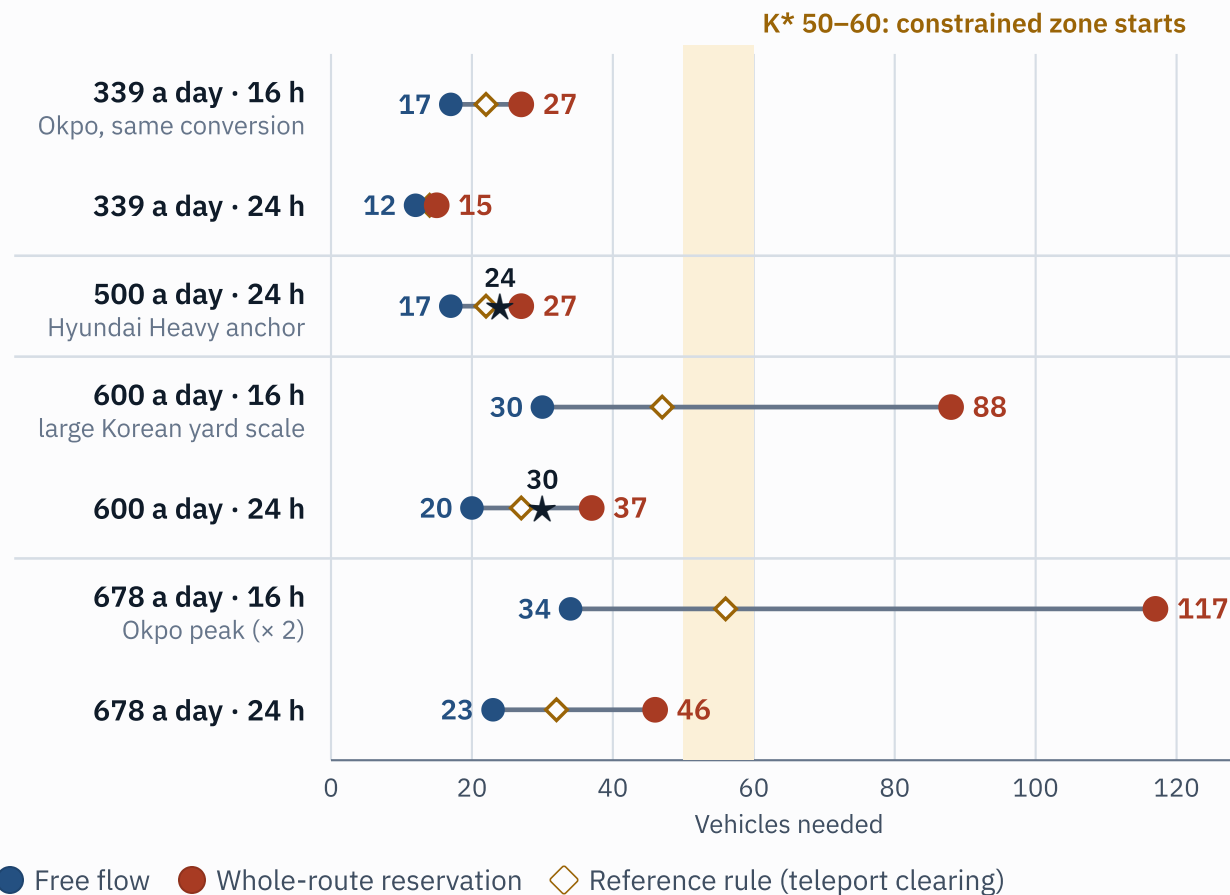
Migration sources (full texts, T24): a railway block section is held by one train (Abril et al. 2008); shorter blocks use less capacity, the best block length is open (Goverde et al. 2013); UIC 406 (Landex et al. 2008); a mine's single lane with passing bays run as block sections (Eustace et al. 2021).

Two examples of segmenting: the main scenario lifts only the upper edge; with the dock binding, both edges rise



Sources: thesis-core a882742, docs/T22_report.md §§3.5, 4; the chart's numbers are read by make_data.py from the batches' curves.csv (whole road 49d2383872e7, dfb2758822ad; segmented 656e2da75368, a1f485302418). Conditions: saturated demand, 4 days, 10 seeds, whole-route reservation; block boundaries 25 m either side of each stop on the dock road (a scenario setting); segmented is Okpo's main scenario (author's imagery check, 9 Oct 2026), the whole road the comparison; Okpo derived map rev 1825. Probe: review appendix A.

Vehicles needed at whole-yard volume: the real scale of “worth so many vehicles”



■ Constrained-zone start K* (50–60) ★ Field fleet

Sources: thesis-core a882742, docs/T22_report.md §§6, 7 and results/t22_report_3287d62c7916/t22_demand.csv (integer K, 10 seeds; vehicles needed = smallest K whose mean throughput $\geq 0.99 \times$ the daily volume; dock road as one resource, segmented within one vehicle); K* interval from t22_summary.json. 339 = Okpo's steel use by the same conversion (yearly 294–363), the Hanwha system's direct anchor 350, peak $\times 2$ (literature library, notes/occupancy_anchors_20261009.md §2.5). Field anchors: Park and Seo 2006; Shen et al. 2018 (full texts, T24). Okpo derived map rev 1825.

17 → 27

Okpo's own 339 a day (T24, same conversion), 16 h: free flow → whole-route reservation; under 24 h, 12 → 15. Both below the constrained-zone start K* (50–60).

34 → 117

Okpo's peak, 678 a day ($\times 2$), 16 h: inside the network-constrained zone; under 24 h, 23 → 46, at the zone's lower edge.

24 · 30

Field fleets: Hyundai Heavy, about 500 moves a day, 24 vehicles, 24 h; a large Korean yard, about 600 a day, about 30 vehicles (shift not stated, read at 24 h). Both sit between free flow and reservation under a 24 h day.

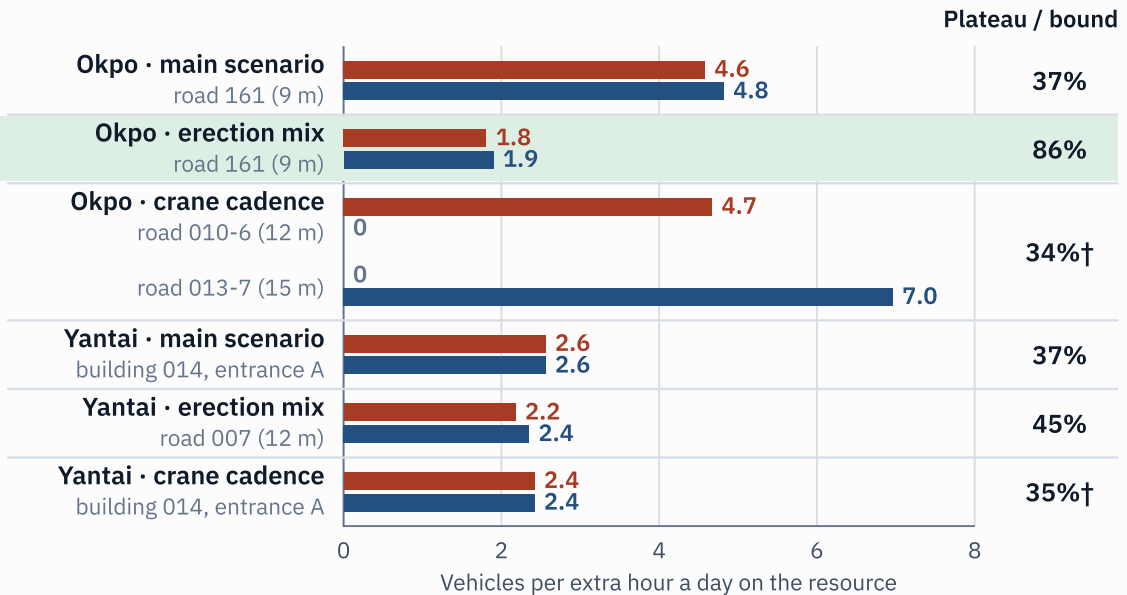
Whether the working day is 16 h or 24 h, and the volume the daily mean or the peak, is H1's main caliber, which the author decides after T27; both are shown here. The exchange rate addresses the network-constrained zone (H4).

T6, the exchange rate: the bound's dual prices in vehicles; only binding elements are worth changing

$$\pi_r = \partial T1' / \partial H_r \quad \text{extra tasks a day per extra hour a day on resource } r$$

$$e_r = \pi_r \div (H / \bar{c}) \quad \text{in vehicles; } H / \bar{c} = \text{one vehicle's daily capacity}$$

- **Only upgrades on binding constraints have value:** a slack resource has a zero dual price.
- **With plateau / bound near 1, e_r is the exchange rate;** far below 1, it bounds an upgrade's value from above.
- **Prices are marginal and the solution degenerate:** Yantai's building 014 has two entrances full at once; changing one adds only 2.3 a day, both together +144 to first order; T25 re-solves combined changes.
- **Two columns side by side:** read the T1' column under shortest-path dispatch and the T1'' column when re-routing is allowed (chapter 4's detour orchestration); new links always use T1''.



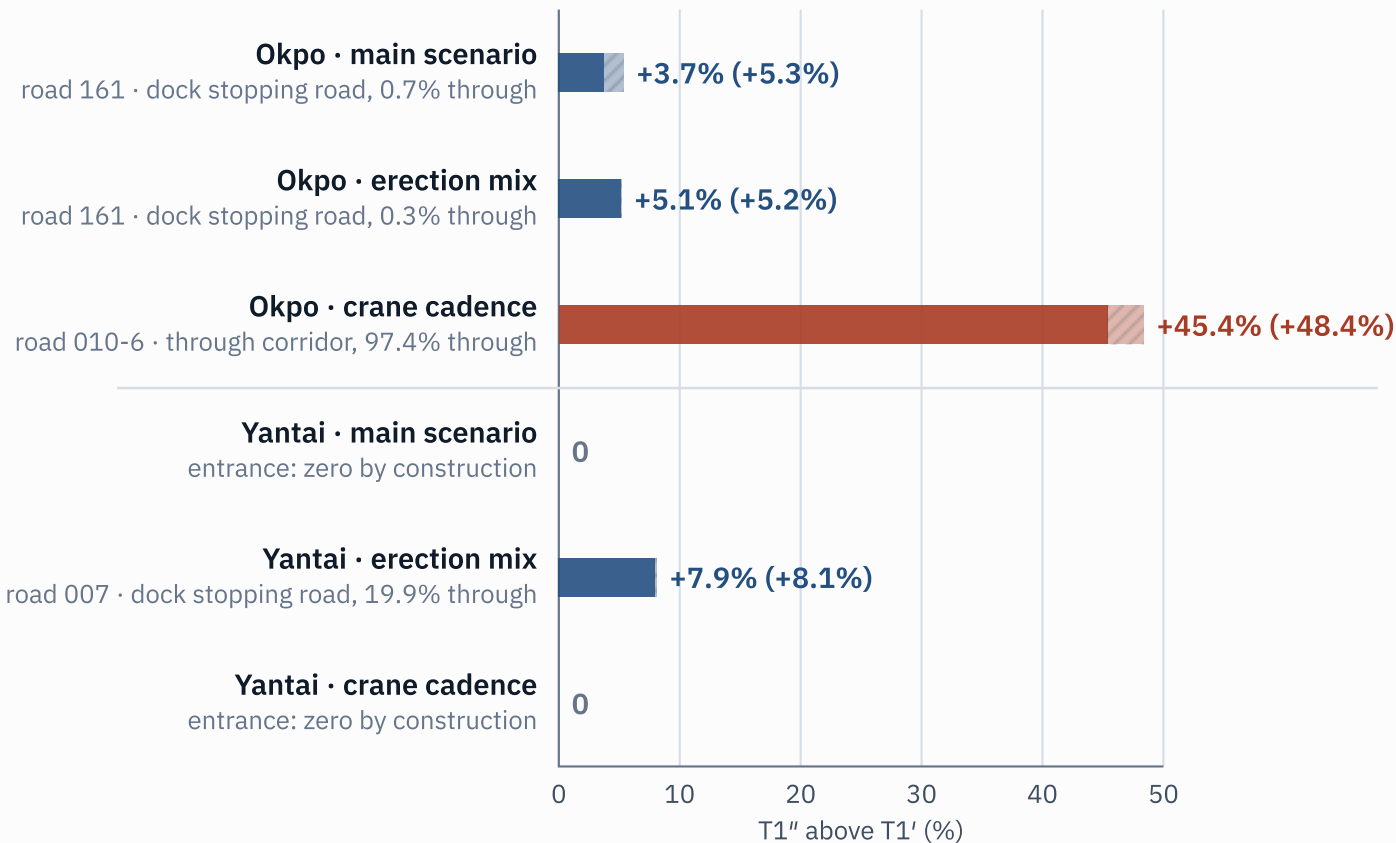
■ Shortest-path dispatch (T1') ■ Re-routing allowed (T1'')

† crane cadence on the system caliber; others on the completed mix

Extra tasks a day per extra hour a day on the resource, divided by one vehicle's daily capacity (about 24 at Okpo, 32 at Yantai). The erection mix reads as the exchange rate; elsewhere the plateau is 36–45% of the bound, so the vehicles bound an upgrade's value from above.

Sources: theorem as stated in the manifesto v1 theorem list, T6 (status: author); thesis-core a882742, prices.csv of results/97d667d16273 and results/e7f94238b96d (system T1' and T1'' columns; resources with a positive price among the top 10 of either) and platform.csv (plateau / bound on the completed-mix caliber; † crane cadence on the system caliber, T23 report §4.4); H / $\bar{c}$ = fleet bound ÷ K in the curves.csv of the scenario's batch; docs/T23_report.md §§1.2, 4. Crane cadence = crane count from the map, P6 fixed in absolute terms (T21). Okpo rev 1825, Yantai rev 396.

Route slack in the bound: the most re-routing can add, and where road changes and orchestration substitute



■ Lower: back-substituted feasible value – T1' ■ Up to T1'' – T1'

Through corridor: road 010-6 (Okpo, crane cadence)

97% of its traffic passes through, and the detour keeps 22% spare at 235 m longer on average. In the T1' column a 10% shorter occupancy is worth +198 a day (≈ 8.3 vehicles); in the T1'' column, 0: re-routing can stand in for changing it.

Dock stopping roads and entrances

Dock stopping roads (Okpo road 161, Yantai road 007) carry little through traffic; their slack comes mainly from entering the stop from the other end: +3.7% to 7.9%. An entrance has zero slack by construction; only the door itself can change (T25, door granularity).

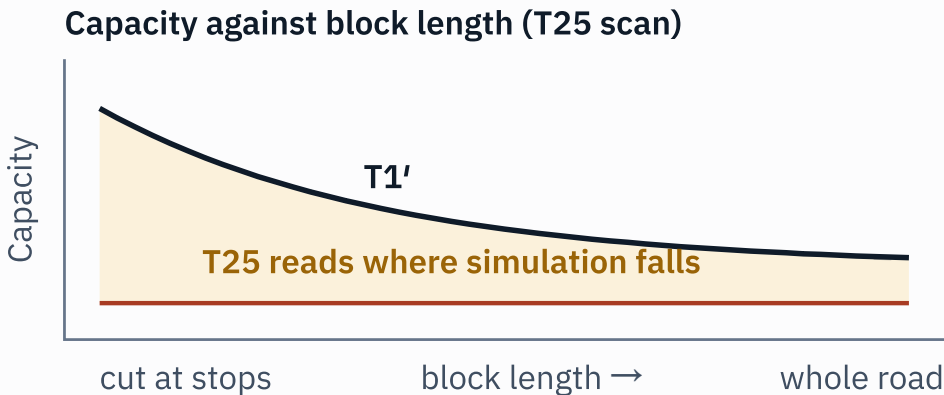
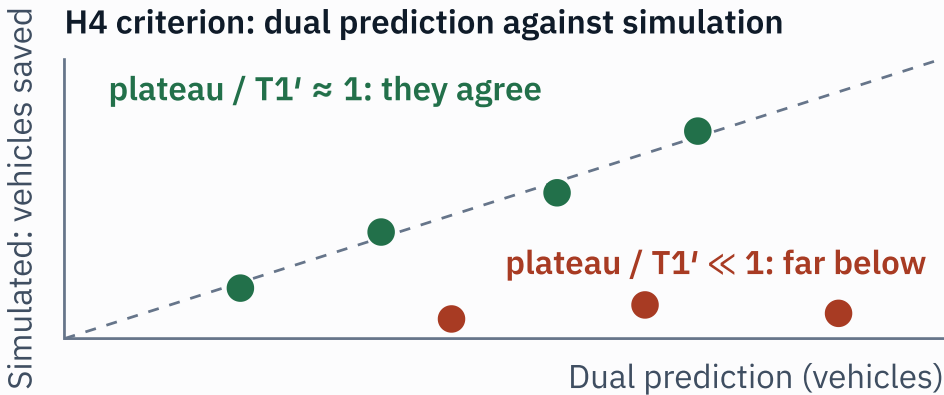
Route slack is a difference of two bounds: the most chapter 4's detour orchestration could recover; what it does recover comes from that orchestration's plateau. The three readings are proposed wording; T25 adds points.

Sources: thesis-core a882742, route_free.csv of results/97d667d16273 and results/e7f94238b96d (lower = back-substituted feasible value – system T1', upper = T1'' – system T1'), readout.csv (through share, detour slack) and prices.csv; docs/T23_report.md §§2.2, 2.6, 4.3; draft proof of T1'' in docs/T23_route_free_proof.md (rewritten by the author), no counterexample found by the independent review (00_总控/reviews/T23.md). Okpo rev 1825, Yantai rev 396.

The first exchange-rate table (T25, after T27): each change in vehicles, dual prediction against simulation

CHANGE	READINGS SO FAR	DUAL PREDICTION (T23)	SIMULATION: Δ VEHICLES NEEDED
Segmented blocking (direction locks)	Erection mix: T1' +76.5%, reservation +8.3% (T22)	Road 161: 1.8 vehicles per hour a day	T25
Dock-front holding bay	—	not seen by the LP	T25 · T26
Passing bays	Probe: −0.2% beyond segmenting	T1'' on the new network	T25
Widening	Probe: 12 m, +4.1%; road 161 +1 m: bound unchanged	Next-step pricing; Yantai road 007 +1 m: +127 a day ≈ 4.0 vehicles	T25
Door granularity (Yantai bldg 014)	—	One entrance +2.3 a day; both, first order +144	T25

T25's caliber: 10 seeds, full K grid; vehicles needed at Okpo's 339 a day and the 678 peak, 16 h and 24 h side by side; on T27's rules and Okpo rev 1833; conclusions only from rules without teleport clearing.



Both figures are schematics, not data

Sources: task board T25 (retrofit simulations and the first exchange-rate table; block-length scan; door granularity; dock three-resource network; after T27, with the map switch moved to T27's step 0); readings so far from docs/T22_report.md §3.5 and review appendix A; dual predictions from docs/T23_report.md §§4.2, 4.3 and §7 item 5 (numbers read by make_data.py); manifesto H4 criterion and wording rule 2 (teleport rule as reference only). The two figures on the right are schematics.

Bilevel co-design: changes and fleet outside, orchestration inside, dual prices to rank

Chapter 5's results ladder

FLOOR

Grid evaluation of fleets and road changes

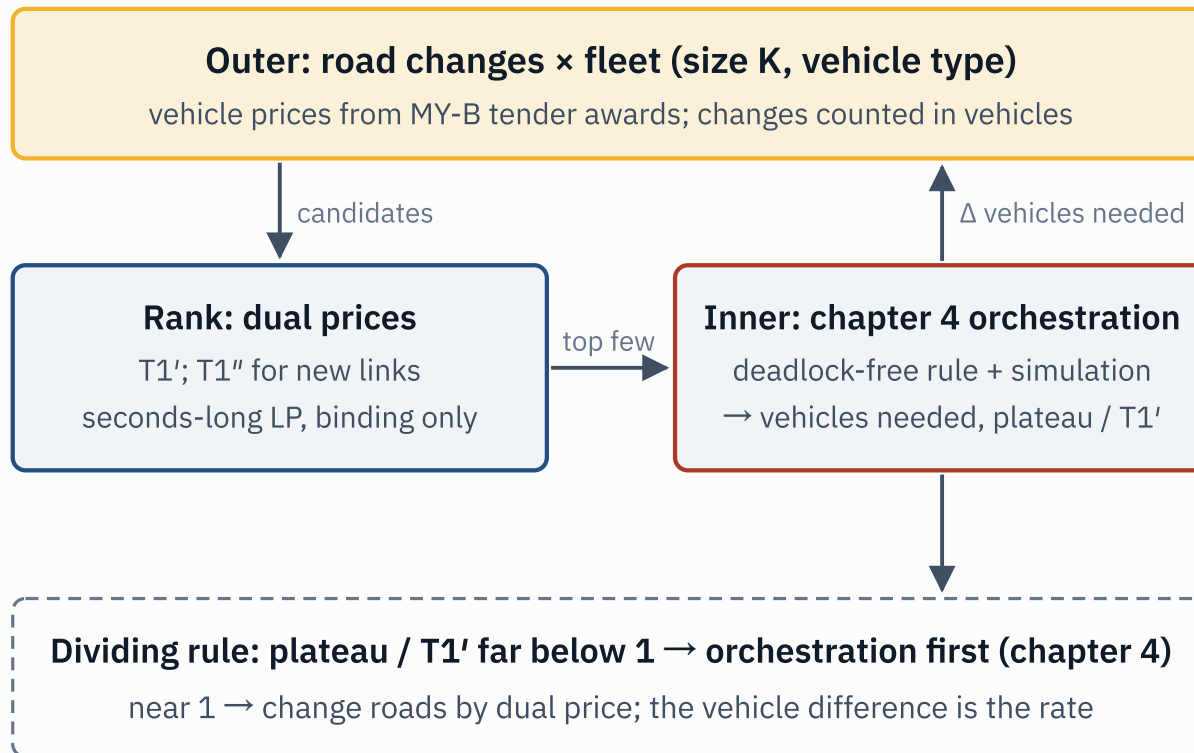
TARGET

The T6 exchange rate and critical-link identification

CEILING

Bilevel co-design with optimality gaps

- Dual prices shrink the outer search to the binding elements, cutting the designs to simulate; T23's bounds and price table take about 6 min for the largest scenario and at most 25 s for the others.
- The inner level currently uses whole-route reservation (the best of the rules T22 tested); T27's less conservative rule replaces it once ready.
- Key-element review is a fixed step: the author checks the top three dual-price elements on the imagery, and the resource granularity is chosen from that before recomputing.



H7, vehicle type and occupancy: large vehicles occupy even when empty; small ones can yield but run extra empty trips to team up

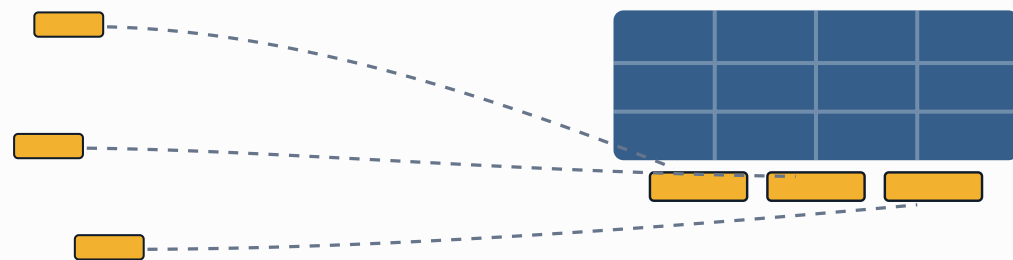
Large vehicle: holds the lane even when empty



Small vehicle: yields where passing is possible when empty



Small vehicles teamed: a giant block needs several, each running empty to meet



- **Where MY-B starts:** with static shortest paths and no traffic, one large enough vehicle is cheaper than any coupled team for every block; prices come from 33 public tender awards. H7 asks whether occupancy changes that answer.
- **Nearest same-field work (full texts):** Tao et al. 2017 compare vehicle types by simulation with road capacity counted in vehicles; Li et al. 2020 model two-vehicle cooperative transport but leave roads out. H7 adds the team's footprint and road holding, the empty runs to meet and the return in convoy.
- **The case (T26, after T25):** Hanwha Ocean's 4Q24 IR lists 3 floating docks (“Floating Dock 3EA”); Okpo lifts giant blocks into them with sea cranes, modelled as a quay-side sink plus a floating-crane resource, with floating-dock flow as a separate scenario. Giant blocks travel on 2–4 coupled vehicles; “about 10 giant blocks per ship” comes from another yard (Koh et al. 2011), not Okpo.

[Open the fleet-sizing \(MY-B\) briefing →](#)

Established, and under way: what each task feeds

Established (T22–T24, passed review 9–10 Oct 2026)

- **Both edges of the capacity interval (T22, 10 seeds):** the best deadlock-free rule is whole-route reservation; Okpo main [627, 1,720], erection mix [611, 703], Yantai [455, 1,252]; the dividing rule has two examples.
- **A bound free of sampling noise, with a route-free version (T23):** system T1' 1,736 at Okpo, 1,298 at Yantai; T1'' carries a certificate independent of solver tolerance, and the independent review found no counterexample.
- **Dual prices in two columns, convertible to vehicles (T23):** one more hour a day on road 161 in the erection mix ≈ 1.8 vehicles; route slack from 0 to +45.4%.
- **External anchors (T24, full texts):** Okpo 339 a day; under a 24 h day field fleets sit between free flow and reservation; the closest same-field work, Tao et al. 2017, compared item by item.

T27

running A less conservative deadlock-free rule (Banker plus a step-wise feasibility search, or claims cut at the next safe holding point, theorem T5); step 0 moves Okpo to rev 1833, fixes the direction-lock U-turn at stops and splits reservation waits. → re-check of the lower edge, H2's decline and H5's cost of safety

T25

after T27 Retrofit simulations; the first exchange-rate table; door granularity; capacity vs block length; the dock three-resource network. → the exchange-rate table, the prediction-vs-simulation figure, the granularity curve

T26

after T25 The floating-dock scenario; finite holding area S and $K_0(S)$. → the H7 case section; the rule “what a holding area is worth in vehicles”

H1

author Decided together after T27: H1 part 2's new numbers, part 3's two-layer wording, and the main caliber (16 h / 24 h, daily mean / peak). → this briefing shows both scales until then

Risks and mitigations

RISK	MITIGATION
In the main scenario road changes barely move throughput	That is itself the C3-side reading: orchestration first; once T27 and chapter 4 raise the lower edge, read the exchange rate again at the new plateau / bound
The best deadlock-free rule may change (T27 running), moving the lower edge and plateau / bound	The boundary and the rate always name the rule; the new rule joins the interval table through the same batches and the ratios are recomputed, while the dividing rule itself stays as stated
The dual is degenerate and marginal; width works in steps	T23 already gives re-solved increments (building 014's two entrances must change together; 1 m more on road 161 gives 0); T25 prices combined changes and “widen to the next step”, relaxing one bottleneck at a time to find the next
Whether the model's vehicle counts match the field	Answered: under a 24 h day Hyundai Heavy's 24 vehicles and about 30 at a large Korean yard both sit between free flow and reservation (T22, T24); the main caliber, 16 h / 24 h and mean / peak, is the author's after T27, and both are reported until then
The 200 m and 25 m of segmenting are scenario settings	Author's imagery check: with a vehicle at the dock on road 161, the neighbouring sites stay reachable; T25's block-length scan turns the 25 m into a capacity–granularity curve
Gaps in public material on floating docks and giant blocks (current dock sizes, convoy speed)	T24 has found the number of floating docks and the sea-crane lifting; floating-dock flow is a separate scenario, with synthetic parts labelled as scenarios

Sources: docs/T22_report.md §§6, 10 and docs/T23_report.md §§4.3, 7 (thesis-core a882742); review records 00_总控/reviews/T22.md, T24.md; task board T25–T27 and the author's imagery check (9 Oct 2026); literature library, notes/occupancy_anchors_20261009.md §5; manifesto wording rules 1, 2, 5.

Target journals (paper 3, chapter 5): only journals in the school's 2026 directory

Is a passing bay worth a transporter? Fleet-network co-design for road-occupying transport

FIRST CHOICE · CLASS B

- *Ocean Engineering* (class B)
- *IEEE Transactions on Automation Science and Engineering* (class B)

Shipyards transport mechanism and engineering gain (OE); network–fleet co-design and the bilevel algorithm (T-ASE)

STRETCH · CLASS A

- *Reliability Engineering & System Safety* (class A)

Condition: written as capacity loss under critical-link failure and redundancy design (the redundancy value of holding areas and second entrances)

ALTERNATIVES · CLASS B

- *Transportmetrica A: Transport Science* (class B)
- *Expert Systems with Applications* (class B)
- *Applied Soft Computing* (class B, bilevel algorithm)
- *Journal of Manufacturing Processes* (class B)

SUPPLEMENTARY · CLASS C

- *Computers & Industrial Engineering* (class C)
- *Journal of Ship Production and Design* (class C)

Class B first; class A as a stretch under the condition on its card. Graduation needs at least one paper in a class B or higher journal, and this paper aims at class B.

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Back to the whole thesis

Is a passing bay worth a transporter? First see how far the plateau sits below the bound.

<< 1

Plateau / T1' far below 1: the rule is the bottleneck; orchestration first.

≈ 1

Plateau / T1' near 1: the bound is tight, dual prices are the exchange rate, and fixing the road pays; afterwards the bottleneck moves to the rule (0.87 → 0.53).

H7

Does occupancy change the choice between large vehicles and teams? Tested on floating-dock transport (T26).

← Back to the thesis briefing

Occupancy model and bounds →

Capacity interval →

Traffic orchestration →

Multi-yard test →

Fleet sizing (MY-B) →

Thank you.